

[22]
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```
import pandas as pd
```

Double-click (or enter) to edit

Double-click (or enter) to edit

[23]
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```
data=pd.read_csv("/content/online_retail_II(1).csv", encoding='iso-8859-1', sep=',', on_bad_lines='skip')
```

[24]
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```
data.head(20)
```



	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
5	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010 08:26	7.65	17850.0	United Kingdom
6	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-2010 08:26	4.25	17850.0	United Kingdom
7	536366	22633	HAND WARMER UNION JACK	6	01-12-2010 08:28	1.85	17850.0	United Kingdom



▼	7	536366	22633	HAND WARMER UNION JACK	6	01-12-2010 08:28	1.85	17850.0	United Kingdom
	8	536366	22632	HAND WARMER RED POLKA DOT	6	01-12-2010 08:28	1.85	17850.0	United Kingdom
	9	536368	22960	JAM MAKING SET WITH JARS	6	01-12-2010 08:34	4.25	13047.0	United Kingdom
	10	536368	22913	RED COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047.0	United Kingdom
	11	536368	22912	YELLOW COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047.0	United Kingdom
	12	536368	22914	BLUE COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047.0	United Kingdom
	13	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010 08:34	1.69	13047.0	United Kingdom
	14	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	01-12-2010 08:34	2.10	13047.0	United Kingdom
	15	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	01-12-2010 08:34	2.10	13047.0	United Kingdom
	16	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-2010 08:34	3.75	13047.0	United Kingdom
	17	536367	22310	IVORY KNITTED MUG COSY	6	01-12-2010 08:34	1.65	13047.0	United Kingdom
	18	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01-12-2010 08:34	4.25	13047.0	United Kingdom
	19	536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-2010 08:34	4.95	13047.0	United Kingdom

[25]

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data.info()



```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 541910 entries, 0 to 541909
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Invoice                541910 non-null object
1   StockCode              541910 non-null object
2   Description            540456 non-null object
3   Quantity              541910 non-null int64
4   InvoiceDate            541910 non-null object
5   Price                 541910 non-null float64
6   Customer ID           406830 non-null float64
7   Country               541910 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
```

[26]

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```
data.isna().sum()
```

...

0

Invoice

0

StockCode

0

Description

1454

Quantity

0

InvoiceDate

0

Price

0

Customer ID

135080

Country

0

dtype: int64

[27]

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data.describe()



	Quantity	Price	Customer ID
count	541910.000000	541910.000000	406830.000000
mean	9.552234	4.611138	15287.684160
std	218.080957	96.759765	1713.603074
min	-80995.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13953.000000
50%	3.000000	2.080000	15152.000000
75%	10.000000	4.130000	16791.000000
max	80995.000000	38970.000000	18287.000000



[29]

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data.shape



(541910, 8)